Fiscal Year:	FY 2015	**Task Last Updated:**	FY 07/23/2014
PI Name:	Grabham, Peter Ph.D.		
Project Title:	Effects of Space Radiation on the Function of Human Brain Capillaries and the Blood-brain-barrier: Implications for Neurodegenerative Disorders		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pwg2@cumc.columbia.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	646-761-1275
Organization Name:	Columbia University		
PI Address 1:	Center for Radiological Research		
PI Address 2:	630W W 168th St, VC 11-243		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2015 per NSSC information (Ed., 8/29/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ponnaiya, Brian (Columbia University) Smilenov, Lubomir (Columbia University)		
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Task Description:	This proposal is aimed at determining the effects of space radiation on the function of human brain blood vessels (capillaries) and the Blood-Brain-Barrier (BBB). The brain vascular system is crucial to healthy functioning of the brain and its dysfunction is not only a primary event in a range of neurodegenerative diseases but also an important influencing factor in many others. Furthermore, vessel pathologies involve common susceptibility factors such as oxidative stress and inflammation so any radiation effects on capillaries can be influenced by other preexisting conditions. Here, we propose to investigate the effects of space radiation on angiogenesis and BBB function using human brain blood vessel cells in 2-dimensional and 3-dimensional tissue models. Since the brain vascular system is essentially an integrated multi-cell system we also propose to utilize other human brain cell types in tissue models to create the most realistic human vessel model possible. We have focused on the main aspects of vessel function that are relevant to the research emphases of acute and late risks to the brain. That is, functioning of the BBB, blood vessel development and the compound effects of preexisting conditions such as inflammation. A correctly functioning blood system supplies the brain and other tissues with oxygen and nutrients while preventing deleterious blood-borne molecules from entering the brain and causing possible damage to brain cells. Tight junctions between the blood vessel cells are formed by proteins that make a tight seal and prevent leakiness. We will determine the effect of space radiation on blood vessels of the brain and its possible impact on neurodegeneration by testing for vessel formation (angiogenesis) vessel leakiness, and the influence of other factors such as inflammation molecules after exposure to both galactic cosmic radiation and radiation from the sun.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gamma radiation studies Studies to optimize assay conditions using gamma radiation revealed a previously unknown effect of gamma photons on human blood vessels. It is the first demonstration of a transient and rapid effect of gamma radiation on human endothelial barriers that involves cell detachment and loss of the cell adhesion molecule - PECAM-1. This effect occurred at the fractionated radiotherapy dose of 2 Gy and therefore has the potential to be of clinically relevant importance considering the role of PECAM-1 in many of the cellular processes involved in pathologies seen at longer times after radiation therapy. As such it could be a target for treatments that alleviate these pathologies. A second potential benefit is that of drug delivery to tumors. A temporary opening of the endothelial barrier by the dual effects of low temperatures and gamma radiation could be a means by which impermeable chemotherapy drugs could be delivered directly to tumors. Tissue models The development of 3-Dimensional human tissue models from normal human cells and stem cells has great potential in many fields of medical research. Tissue models can more accurately depict human tissue since the cells can be arranged spatially as they would be in vivo and can interact with each other as they would in the human body. A neurovascular unit can be used for basic research on many aspects of the human brain. These include regeneration, synaptic function, and degeneration. Because the tissue model is derived from individual cells, each cell type can be altered genetically before it is incorporated into the model.
Task Progress:	Task progress Endothelial barrier studies Protons. Experiments on the effect of protons on barrier function are now almost completed. Analysis of all endpoints was carried out for up to several days as in many other studies of endothelial barrier function in mainstream vascular biology. We have carried out studies with high-energy (1 GeV) protons. Our results to date suggest that protons do not appear to disrupt the endothelial barrier up to relatively high doses. 4 Gy did not significantly affect the endothelial barrier up to 72 hours after exposure. Trans-endothelial resistance, permeability, and the presence of tight junctions and cell adhesion molecules showed little change. It appears that low LET protons, like gamma rays, might only affect the endothelial barrier by doses high enough to cause cytotoxic effects. If this is the case, then low LET protons would not constitute a hazard in the space environment. This work together with comparisons with the effects gamma rays will be the basis for another paper in a radiation journal. Fe ions. We have investigated the acute effects of Fe particles on human endothelial barrier models and determined that these charged particles have detrimental effects on barrier function. All endpoints – TEER, permeability, and tight junction integrity in monolayers and in 3-D vessel models, showed changes associated with loss of function at a dose (0.50 Gy) that has also previously been shown to degrade the structure of vessel models. However, effects on permeability and tight junction integrity were also compromised at a lower dose of 0.25 Gy. This indicates that endothelial barrier function can be disrupted without breakdown of the tissue. Thus a morphologically intact vessel can still sustain enough damage from a lower dose to cause dysfunction. The dose for this response is close to the dose for a Mars trip and taking into account the physical properties of the particle, the biology of the response and the nature of the progression of degenerative diseases, this effect has the potential to be a health concern in space travel. This work was published in the new journal – Life Sciences in Space Research (Sharma et al., 2014). Angiogenesis studies We have used assays that have enabled us to distinguish between the inhibition of an early stage and a late stage of vasculogenesis. The unexpected high relative biological effectiveness of low LET protons compared to high LET Fe ions on vessel formation, can be explained by the fact that particle radiation can inhibit vasculogenesis in two distinct ways according to the LET of the radiation. Low LET protons, which deposit energy more evenly throughout the cell, inhibit the early stages during the extension of motile tips and the formation of guidance tunnels. High LET Fe ions do not inhibit the early stages only the later stages of tube formation. These results have implications for both human space travel and for particle radiotherapy. This work has now been published in Vascular Cell (Grabham et al., 2013) along with a review (Grabham and Sharma 2013) in the same journal. We have also completed studies to determine the LET ranges for each type of inhibition. Using the assays described in the previous section we have tested the effects of several particles and energies to give a range of LETs between 0.22 (1GeV protons) and 150 (1GeV Fe ions). Particles with an LET of 8 KeV/n and above inhibit the later stages of angiogenesis while particles with an LET of 2.9 KeV/n and below inhibit the early motility stages of angiogenesis. This

	work is being prepared as a manuscript.
Bibliography Type:	Description: (Last Updated: 03/04/2024)
Abstracts for Journals and Proceedings	Grabham P, Sharma P, Bigelow A, Geard C. "Distinct mechanisms of the inhibition of vasculogenesis by different species of ionizing particles." Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. J Radiat Res. 2014 Mar;55(Suppl 1):i44-i45. (Proceedings of Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013.) http://dx.doi.org/10.1093/jrr/rrt172 ; PMCID: PMC3941500 , Mar-2014
Articles in Peer-reviewed Journals	Grabham P, Sharma P, Bigelow A, Geard C. "Two distinct types of the inhibition of vasculogenesis by different species of charged particles." Vascular Cell. 2013 Sep 17;5(1):16. http://dx.doi.org/10.1186/2045-824X-5-16 ; PubMed PMID: 24044765; PubMed Central PMCID: PMC3856512 , Sep-2013
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